

Wizard Optimization Math

Core formulas

Spell Save DC = 8 + INT mod + PB | Spell Attack = INT mod + PB
Concentration DC = max(10, floor(damage / 2)), maximum DC 30

1. Concentration Survival

Cells show **Normal / Advantage**. Advantage uses two d20s and keeps the higher. The repeated-hit table shows the chance that Concentration is still active after every listed DC 10 save succeeds.

DC	+2 CON save	+5 CON save	+8 CON save
10	65.0% / 87.8%	80.0% / 96.0%	95.0% / 99.8%
15	40.0% / 64.0%	55.0% / 79.8%	70.0% / 91.0%
20	15.0% / 27.8%	30.0% / 51.0%	45.0% / 69.8%

CON save	1 hit	3 hits	5 hits	10 hits
+2 N	65.0%	27.5%	11.6%	1.3%
+2 Adv	87.8%	67.6%	52.0%	27.1%
+5 N	80.0%	51.2%	32.8%	10.7%
+5 Adv	96.0%	88.5%	81.5%	66.5%
+8 N	95.0%	85.7%	77.4%	59.9%
+8 Adv	99.8%	99.3%	98.8%	97.5%

2. Spell Save Failure - Example DC 17

Failure chance = chance the target rolls below your Spell Save DC after modifiers. Saving throws do **not** automatically fail on a natural 1 or succeed on a natural 20.

Target save	Normal fail	Target Disadv.	Target Adv.
-2	90%	99%	81%
+0	80%	96%	64%
+2	70%	91%	49%
+4	60%	84%	36%
+6	50%	75%	25%
+8	40%	64%	16%
+10	30%	51%	9%

3. Spell Attack Accuracy

Attack	AC 12	15	18	21
+5	70%	55%	40%	25%
+8	85%	70%	55%	40%
+11	95%	85%	70%	55%

Attack rolls always miss on natural 1 and hit on natural 20. Adv hit = 1 - (1 - p)^2. Disadv hit = p^2.

4. Initiative Probability

Your bonus edge	Act first*
-5	28.1%
-2	40.5%
+0	50.0%
+2	59.5%
+5	71.9%
+8	82.0%
Equal + Advantage	66.6%
Equal + Disadv.	33.4%

*Calculator treats ties as 50/50. Actual PC-monster ties are decided by the DM.

Damage & Travel Math

Expected damage, AoE scaling, travel speed and forced-march risk

5. Expected Damage

Save for half: Expected per target = Avg Damage x (0.5 + 0.5 x Fail Chance). **No damage on success:** Avg Damage x Fail Chance. **Spell attack:** E = Hit% x normal damage + Crit% x extra damage dice.

Target fail rate	1 target	2	3	4	5	6
40%	19.6	39.2	58.8	78.4	98.0	117.6
60%	22.4	44.8	67.2	89.6	112.0	134.4
80%	25.2	50.4	75.6	100.8	126.0	151.2

AoE break-even example: 8d6 averages 28 damage (Fireball baseline). At a 60% failed-save rate, 3 targets are worth about 67.2 expected total damage; 5 targets about 112.0. This ignores resistance, immunity, overkill, friendly fire, and non-damage riders.

6. Travel Calculator - 2024 Pace

Pace	Miles / hour	Miles / 8h	Travel effect
Fast	4	30	Disadv. on Perception/Survival and Stealth
Normal	3	24	Disadv. on Stealth
Slow	2	18	Adv. on Perception/Survival

Mounted burst: riding horses or other mounts can cover **twice the normal distance for 1 hour**; the mounts then need a Short or Long Rest before doing so again.

Special Sustained Speed

Speed	Normal 8h	Fast	Slow
30 ft	24 mi	32 mi	16 mi
40 ft	32 mi	42 mi	21 mi
60 ft	48 mi	64 mi	32 mi
100 ft	80 mi	106 mi	53 mi

DM Toolbox formula: mph = Speed / 10. Fast = normal x 1 1/3; Slow = normal x 2/3; round down.

Phantom Steed

Maintained time	Distance
1 hour	13 mi
2 hours	26 mi
4 hours	52 mi
8 hours	104 mi

The spell explicitly says **13 miles in an hour**, so this specific rate overrides the generic 100-ft Speed conversion. Distances beyond 1 hour assume successful recasting/replacement logistics.

7. Forced March - Cumulative Exhaustion Risk

Beyond 8 travel hours, each extra hour requires a Constitution save: DC 11, 12, 13, 14, 15... Failure gives 1 Exhaustion level. Table = chance of **at least one failure** by that point.

CON save	+1 hour	+2	+3	+4	+5
+2	40.0%	67.0%	83.5%	92.6%	97.0%
+5	25.0%	47.5%	65.9%	79.5%	88.7%
+8	10.0%	23.5%	38.8%	54.1%	67.9%

Example: with +5 CON, pushing 3 extra hours means saves against DC 11, 12 and 13; cumulative risk of at least one Exhaustion level is about 65.9%.

Wizard Economics

8. Spellbook Economics

2024 Wizard rule: copying a newly found level 1+ Wizard spell costs **50 GP and 2 hours per spell level**. Copying a spell you already know into another book costs **10 GP and 1 hour per spell level**.

Spell level	New GP	New hours	Backup GP	Backup hours
1	50	2	10	1
2	100	4	20	2
3	150	6	30	3
4	200	8	40	4
5	250	10	50	5
6	300	12	60	6
7	350	14	70	7
8	400	16	80	8
9	450	18	90	9

Total spell levels	New copy	Backup copy
10	500 GP / 20 h	100 GP / 10 h
25	1,250 GP / 50 h	250 GP / 25 h
45 (one spell 1-9)	2,250 GP / 90 h	450 GP / 45 h
50	2,500 GP / 100 h	500 GP / 50 h
100	5,000 GP / 200 h	1,000 GP / 100 h

Optimizer takeaway

Track **total spell levels**, not just number of spells. A level 9 spell costs nine times as much time and GP to copy as a level 1 spell. Backup books are dramatically cheaper: 20% of the GP and 50% of the time of learning a new spell.

9. Planar Binding Economics

Planar Binding consumes a jewel worth **1,000+ GP**. Higher-level slots extend duration without increasing that listed jewel cost. The table uses the minimum 1,000 GP jewel and a 366-day comparison horizon.

Slot	Duration	Min GP / day	Casts / 366d	Min jewel GP / 366d
5	24 h	1,000.00	366	366,000
6	10 d	100.00	37	37,000
7	30 d	33.33	13	13,000
8	180 d	5.56	3	3,000
9	366 d	2.73	1	1,000

Slot jump	Duration gain	GP/day reduction
5 -> 6	1 d -> 10 d	1,000 -> 100
6 -> 7	10 d -> 30 d	100 -> 33.33
7 -> 8	30 d -> 180 d	33.33 -> 5.56
8 -> 9	180 d -> 366 d	5.56 -> 2.73

Common setup cost: Planar Binding has a 1-hour casting time and the target must remain within 60 ft for the entire casting. The spell text notes an inverted Magic Circle is a typical way to hold the creature. A fresh 2024 Magic Circle consumes 100+ GP of material, so a setup that requires one is at least 1,100 GP before other summoning costs.

Attunement Comparison

A reusable decision framework for Wizard optimization

10. Magic Item Attunement Comparison

2024 rule: a creature can be attuned to no more than **three magic items** at a time. The score below is an **optimizer heuristic, not a game rule**. Rate each category 0-5, multiply by its weight, then add.

Category	Weight	0-5 rating asks...
Core casting	x5	Does it improve spell DC, spell attacks, spell access, or core casting reliability?
Survival / Concentration	x5	AC, saves, concentration, damage avoidance, defensive reactions.
Resource expansion	x4	Extra slots, charges, stored spells, repeatable high-value effects.
Action economy / Initiative	x3	Earlier turns, reactions, bonus actions, extra meaningful actions.
Mobility / Positioning	x2	Flight, teleportation, escape, range control, terrain bypass.
Campaign utility	x1	Information, social, exploration, niche or campaign-specific leverage.

100-point score
Total = 5 x Casting + 5 x Survival + 4 x Resources + 3 x Action + 2 x Mobility + 1 x Utility. Each category is rated 0-5, so the maximum is 100. Use the score to expose tradeoffs, not to replace judgment.

Replacement delta	Interpretation
+10 or more	Strong upgrade - usually worth the slot.
+5 to +9	Good upgrade - check encounter/campaign needs.
0 to +4	Sidegrade - swap only for a specific purpose.
Below 0	Usually keep the current item unless the niche matters more.

Verified 2024 rules basis
D&D; Beyond Basic Rules: **Rules Glossary** (Attunement) and related 2024 rules references used throughout this document. The attunement weights and 100-point scoring system are a custom optimization heuristic.